Project Name

Uber Request Data Analysis using Excel, MySQL and Pandas

Project Type - Exploratory Data Analysis (EDA)

Contribution - Individual

Project Summary -

The Uber Request Data Analysis project focuses on analyzing ride request patterns, ride completion status, cancellation trends, and vehicle availability using various data analytics tools and techniques. The main objective of the project is to identify operational gaps and demand-supply mismatches in Uber ride services.

The project was completed using Microsoft Excel, MySQL, Python libraries such as Pandas, NumPy, and Matplotlib, along with Google Colab for Exploratory Data Analysis (EDA). Initially, the raw dataset was cleaned and preprocessed using Microsoft Excel. Missing values, duplicate records, inconsistent formats, and unnecessary data entries were handled to ensure data quality and reliability.

After cleaning the dataset, dashboards were created in Excel to visually represent important business insights such as ride status distribution, peak request hours, pickup point analysis, and cancellation patterns. These dashboards help in understanding customer demand and operational bottlenecks effectively.

MySQL was used to perform structured data analysis and generate business insights using SQL queries. Queries were written to calculate ride completion rates, identify cancellation hotspots, analyze no-car availability scenarios, and determine peak demand periods.

Further exploratory data analysis was conducted using Python and Pandas in Google Colab. Various visualizations such as bar charts, line graphs, and pie charts were generated to identify trends and relationships in the dataset. Feature engineering techniques such as extracting request hours and time slots were also performed.

The analysis revealed that the highest ride requests occur during morning and evening peak hours. Airport pickup points experienced a higher number of “No Cars Available” cases, indicating supply shortages during high-demand periods. Additionally, cancellation rates increased significantly during rush hours.

Overall, this project demonstrates practical implementation of data cleaning, SQL analysis, dashboard creation, and exploratory data analysis techniques for solving real-world business problems in the transportation domain.

GitHub Link -

https://github.com/pravatsahu05/Uber-Request-Analysis

Problem Statement

Uber experiences frequent demand-supply mismatches during peak hours which results in ride cancellations and “No Cars Available” situations. These operational inefficiencies negatively impact customer satisfaction and service reliability.

The company needs to analyze ride request data to identify patterns related to peak demand timings, cancellation behavior, and vehicle shortages. Through data analysis, meaningful insights can be generated to improve resource allocation and ride availability.

Define Your Business Objective?

The main business objective of this project is to analyze Uber ride request data to identify operational issues and improve customer service efficiency. The project aims to:

- Identify peak ride request hours
- Analyze ride cancellation trends
- Detect areas with high no-car availability
- Improve understanding of demand-supply gaps
- Generate actionable insights for better fleet management

General Guidelines : -

- Well-structured data cleaning process
- Proper SQL query analysis
- Effective visualizations and dashboards
- Meaningful business insights
- Clear documentation and explanations

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1. Know Your Data

Import Libraries

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

Dataset Loading

The Uber Request dataset was loaded into Python using Pandas for analysis.

df = pd.read_csv("uber_request_cleaned_data.csv")

Dataset First View

df.head()

The dataset contains information regarding:

- Request ID
- Pickup Point
- Driver ID
- Ride Status
- Request Timestamp
- Drop Timestamp
- Request Hour
- Time Slot

Dataset Rows & Columns Count

df.shape

The dataset contains multiple records representing Uber ride requests.

Dataset Information

df.info()

The dataset includes both numerical and categorical variables.

Duplicate Values

df.duplicated().sum()

Duplicate rows were checked and removed during preprocessing.

Missing Values/Null Values

df.isnull().sum()

Missing values were identified and handled appropriately.

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2. Understanding Variables

Categorical Variables

- Pickup Point
- Status
- Time Slot

Numerical Variables

- Request Hour
- Request ID

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3. Data Cleaning and Preprocessing

The following preprocessing steps were performed:

- Removed duplicate records
- Handled missing values
- Standardized date and time formats
- Created additional columns such as:
  - Request Hour
  - Request Date
  - Time Slot

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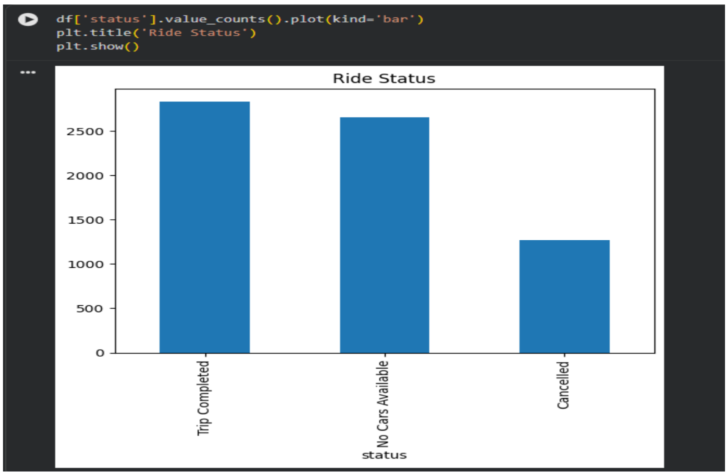
4. Data Visualization and Analysis

Ride Status Distribution

A bar chart was created to analyze ride completion, cancellations, and no-car availability.

Observation

Trip Completed rides were highest, while “No Cars Available” cases were significantly high during peak hours.

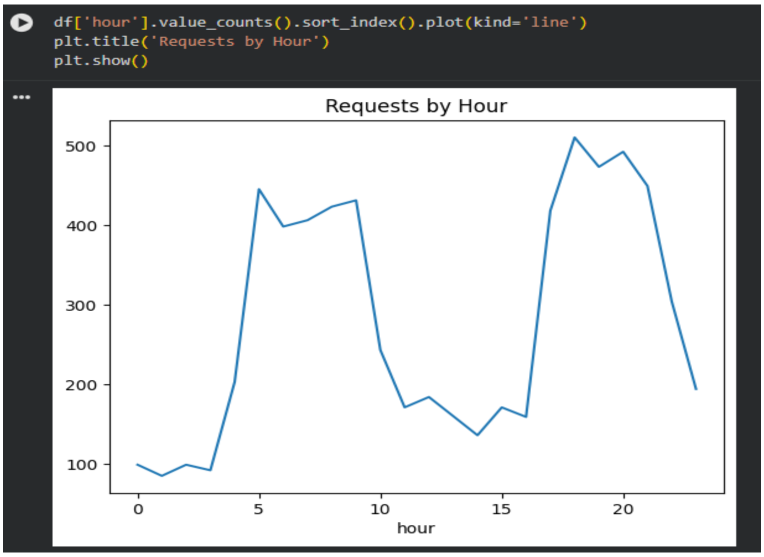


Requests by Hour

A line chart was generated to identify peak demand timings.

Observation

Ride requests were highest during morning and evening hours.

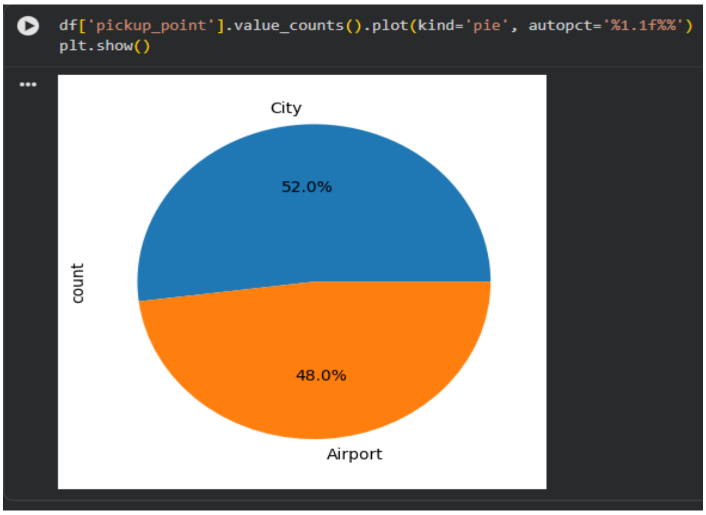


Pickup Point Analysis

A pie chart was used to compare Airport and City ride requests.

Observation

Airport requests showed higher operational inefficiencies due to vehicle shortages.



Time Slot Analysis

Ride requests were grouped according to time slots such as Morning, Afternoon, Evening, and Night.

Observation

Evening and Morning slots recorded maximum demand.

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5. SQL Analysis

MySQL was used to generate insights from the dataset.

Queries Performed

- Ride Status Distribution
- Peak Request Hours
- Cancellation Analysis
- Pickup Point Analysis
- No Cars Available Analysis
- Completion Rate Analysis

Sample SQL Query

```
SELECT status, COUNT(*) AS total

FROM uber_requests

GROUP BY status;
```

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6. Key Insights

- Peak ride demand occurs during morning and evening hours.
- Airport pickup points face higher “No Cars Available” cases.
- Cancellation rates increase during rush hours.
- Demand exceeds supply during peak timings.
- Ride completion rate remains high overall.

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7. Conclusion

The Uber Request Data Analysis project successfully identified operational inefficiencies and demand-supply gaps using data analytics techniques. Through Excel dashboards, SQL analysis, and Python-based EDA, meaningful business insights were generated regarding ride demand, cancellation trends, and vehicle availability.

The project demonstrates practical implementation of data analytics concepts including data cleaning, visualization, SQL querying, and exploratory data analysis. These insights can help transportation companies improve fleet management, reduce cancellations, and enhance customer satisfaction.

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8. Recommendations

- Increase driver availability during peak hours
- Improve airport fleet allocation
- Implement surge-based driver distribution
- Reduce cancellation rates through optimized scheduling
- Use predictive analytics for future demand forecasting

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9. Tools and Technologies Used

Tool/Technology	Purpose
Microsoft Excel	Data Cleaning & Dashboard
MySQL	SQL Analysis
Python	Exploratory Data Analysis
Pandas	Data Manipulation
Matplotlib	Data Visualization
Google Colab	Notebook Environment
GitHub	Project Hosting

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10. References

- Uber Request Dataset
- Pandas Documentation
- Matplotlib Documentation
- MySQL Documentation
- Microsoft Excel Documentation
